INCIDENT REPORT

Meridian Petrochemical Plant

Report Number:	IR-2024-0847
Date of Incident:	2024-03-15
Asset:	Pump P-101 (AquaFlow AF-3500C)
Location:	Production Unit 3
Severity:	HIGH -- Unplanned Shutdown
Reported By:	S. Patel, Shift Supervisor
Downtime:	18 hours

1. Incident Description

At approximately 14:30 on March 15, 2024, Pump P-101 experienced an unexpected bearing failure during normal operation. The control room received a high vibration alarm at 14:22, followed by a high bearing temperature alarm at 14:25. Vibration levels reached 8.2 mm/s RMS, significantly above the 4.5 mm/s alarm threshold. Bearing temperature reached 112 degrees Celsius before automatic shutdown was triggered at 14:30.

The pump was operating at normal conditions: 2950 RPM, 145 cubic meters per hour flow rate, inlet pressure 3.2 bar, outlet pressure 9.8 bar. No abnormal conditions were recorded in the preceding 24 hours. The last scheduled bearing inspection was performed 90 days prior, on December 15, 2023, at which time no defects were noted.

2. Root Cause Analysis

The drive-end SKF 6205-2RS bearing was found to have seized due to inadequate lubrication. Upon disassembly, the bearing grease was found to be severely degraded and contaminated with fine metal particles. The bearing inner race showed signs of spalling and surface fatigue. The bearing cage was partially collapsed.

Root Cause: The bearing lubrication degraded faster than expected due to operating temperatures consistently at the upper end of the normal range (72-78 degrees Celsius average, versus the design assumption of 55-65 degrees Celsius). This accelerated grease breakdown and reduced the effective lubrication interval from the OEM-recommended 180 days to approximately 90 days.

Contributing Factor: The OEM maintenance manual (Revision 2.4) specifies a semi-annual (180-day) bearing replacement interval. This interval assumes standard operating temperatures of 55-65 degrees Celsius. The actual operating temperature of Pump P-101 is consistently 10-15 degrees higher due to the higher-viscosity feedstock processed at Meridian. This discrepancy was not identified during the initial commissioning review.

3. Immediate Resolution

Emergency bearing replacement was performed on both drive-end and non-drive-end bearings. New SKF 6205-2RS bearings were installed with fresh Shell Gadus S2 V220 grease. Shaft runout was measured at 0.03 mm TIR (within tolerance). Alignment was verified using laser alignment tool (angular: 0.02 mm, offset: 0.04 mm, both within tolerance). Pump was returned to service on March 16, 2024 at 08:45.

4. Corrective Actions and Recommendations

Action 1 (CRITICAL): Reduce bearing inspection interval from 180 days to 90 days for Pump P-101. This accounts for the elevated operating temperature that accelerates grease degradation. Apply this change to all similar pumps operating above 70 degrees Celsius bearing temperature.

Action 2: Implement continuous bearing vibration monitoring using permanently installed accelerometers. Set alarm threshold at 4.0 mm/s (reduced from current 4.5 mm/s) and trip threshold at 7.0 mm/s.

Action 3: Evaluate transition to a high-temperature grease (Shell Gadus S3 V460XD or equivalent) rated for continuous operation above 80 degrees Celsius. Conduct a 90-day trial starting Q2 2024.

Action 4: Review OEM maintenance schedule for all centrifugal pumps in the plant. Adjust maintenance intervals based on actual operating conditions rather than generic OEM recommendations.

5. Impact Assessment

Total unplanned downtime: 18 hours. Estimated production loss: 2,700 cubic meters of feedstock throughput. Estimated cost: bearing replacement $2,400, labor $3,600, production loss $85,000. Total incident cost: approximately $91,000.

If the bearing inspection interval had been 90 days instead of 180 days, this failure would likely have been detected during a scheduled inspection and addressed as a planned maintenance activity with zero unplanned downtime.